Highlights

CR-SHE: Oblivious homomorphic aggregation for collusion-resilient search and analytics over encrypted multi-cloud IoT data

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- Oblivious homomorphic aggregation hides the matched record set from every provider.
- Search and analytics stay oblivious under $t = n - 1$ collusion, with no server link.
- A precomputed-column fast path makes single-keyword aggregates flat in record count.
- A one-time-pad index mask removes posting-list-length leakage from the stored index.
- 39 ms per aggregate on 405,184 IoT records, 617× below a homomorphic scan.